

MINI-PROJECT REPORT

Model-Based Simulation & Performance Analysis of a Hybrid Electric Vehicle

Executed via MATLAB / Simulink Powertrain Blockset environment

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1. Introduction & Project Background

The global automotive sector is undergoing a rapid paradigm shift from traditional internal combustion engine (ICE) systems toward electrified drivetrains to counteract environmental degradation, resource depletion, and stringent legislative mandates regarding emissions. Hybrid Electric Vehicles (HEVs) combine an internal combustion engine with an electric propulsion system, striking an optimal balance between the extended driving range of conventional petroleum fuels and the localized zero-emission operation of electric motors.

As part of the professional internship requirements at **EdiGlobe** under the Hybrid & Electric Vehicles domain, this technical mini-project focuses on the structural construction, integration, execution, and subsequent system-level verification of an HEV powertrain configuration. To adhere strictly to cost-effective, zero-overhead development metrics, a virtual vehicle deployment approach was implemented using the high-fidelity **MATLAB/Simulink Powertrain Blockset** environment.

2. System Architecture & Component Mapping

The complete digital vehicle setup architecture relies on an integrated multi-domain computational loop. The overarching workspace architecture segregates the complex mathematical operations into highly isolated, clean functional blocks:

- **Scenarios:** The primary input node containing pre-defined speed trajectories, drive-cycle limits, and temporal route demands. For this study, an extended dynamic driving profile spanning $T = 2474$ seconds was successfully loaded.
- **Driver Commands:** Mimics an intelligent human driver utilizing an active Longitudinal Driver closed-loop controller. It acts as an automated tracking system, dynamically modulating acceleration and deceleration commands to close the error margin between the actual speed profile and target metrics.
- **Controllers:** The fundamental processing core or Electronic Control Unit (ECU). This module maps the central Energy Management Strategy (EMS), continuously interpreting battery metrics and operating boundaries to route power vectors cleanly.
- **Vehicle Block:** A combined multi-physics plant block aggregating physical subsystems. This contains the high-voltage Lithium-Ion battery pack model, an electric motor-generator assembly, a standard internal combustion engine, torque-coupling transaxles, and the final 1-DOF longitudinal vehicle chassis body experiencing aero-drag and tire-friction loads.

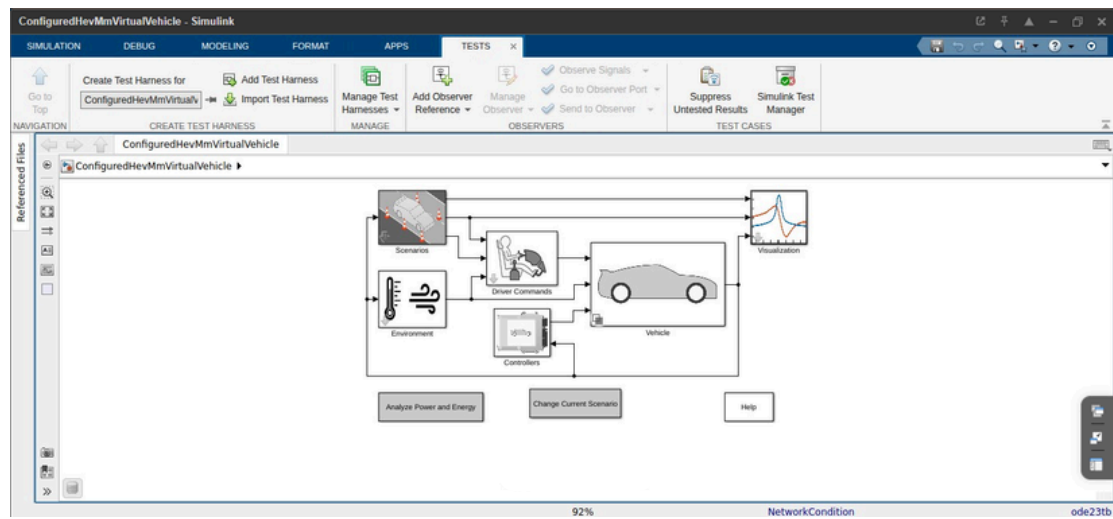


Figure 2.1: Master System Block Layout of the Configured Virtual Vehicle Model in Simulink.

3. Mathematical Foundations & Control Logic

The vehicle's instantaneous mechanical power requirement at the drive wheels, P_{wheel} , is calculated using classical vehicle dynamics equations to overcome aerodynamic drag, rolling resistance, and inertial mass forces:

$$P_{wheel} = (M \cdot a + \frac{1}{2} \cdot \rho \cdot C_d \cdot A \cdot v^2 + M \cdot g \cdot f_r) \cdot v$$

Where M is total vehicle mass, a is linear acceleration, ρ is ambient air density, C_d is the drag coefficient, A is frontal cross-sectional area, f_r is the tire rolling resistance coefficient, and v is the instantaneous forward linear velocity.

The supervisory Control Logic tracks the continuous State of Charge (SoC) of the energy storage system. The rule-based Energy Management Strategy utilizes a hysteresis loop to maintain the optimum state transitions between purely electric vehicle (EV) operation and parallel hybrid modes:

Energy Management Logic Constraints:

- **Condition A (Charge Depleting / EV Mode):** If $SoC > SoC_{min}$ (set at 30%), the system prioritizes purely electric propulsion utilizing the Motor-Generator unit.
- **Condition B (Charge Sustaining / Hybrid Mode):** If $SoC \leq SoC_{min}$, the ECU initiates a command parameter sequence to engage the Internal Combustion Engine to both meet traction demands and drive the generator to recover battery charge.

4. Results, Simulation Analysis & Discussion

The system-level simulation was executed across a comprehensive, highly transient driving environment over a total duration of $T = 2474.00$ seconds. The dynamic system response outputs were tracked using the integrated multi-channel *Performance and FE Scope* cockpit environment.



Figure 4.1: Logged Waveform Trajectories from the "Performance and FE Scope" Dashboard.

4.1 Velocity and Trajectory Coherence

As tracked in the *Trace Velocity* plot, the actual simulated velocity line fits tightly with the reference target trace, maintaining peak speeds near 55–60 mph. This indicates excellent operational integrity of the PID tuning loops within the automated longitudinal driver subsystem.

4.2 Hybrid Component Interactions

The coupled interactions between engine and motor speed (*EngSpd*, *MotSpd* in RPM) and torque (*EngTrq*, *MotTrq* in Nm) show clear alternating patterns. During peak acceleration spikes, the electric motor supplements the engine torque output smoothly. During braking maneuvers, the motor torque drops below zero into the negative quadrant, indicating that kinetic braking forces are being successfully converted back into electricity via regenerative braking mechanisms.

4.3 Battery State and Fuel Optimization

The *Battery Current* graph exhibits dense high-frequency yellow spikes alternating between positive (discharging) and negative (charging) cycles. Thanks to this continuous charge capture via regenerative braking and targeted engine-generator loading, the *Battery SOC* trace maintains a highly optimized, safe equilibrium curve. Consequently, the vehicle achieves an average stabilized fuel economy curve converging cleanly above **40 MPGe**, proving substantial energy conservation.

5. Project Conclusions

This model-based mini-project successfully validates the role of software-in-the-loop (SIL) simulations in verifying automotive powertrain architectures. By deploying a comprehensive HEV plant within MATLAB/ Simulink, high-level control logic and power split strategies were tested under real-world transient loads with zero physical prototype overhead costs. The results conclusively demonstrate that optimizing energy routing between the internal combustion engine and electric motor yields stable battery tracking alongside excellent fuel economy figures.

6. References & Framework Citations

- MathWorks, "Powertrain Blockset Reference Documentation and Preconfigured HEV Models," 2025.
- EdiGlobe Internship Hybrid & Electric Vehicles Domain Learning Modules/Notes, Technical Training Dataset.